

Should we roll out the zone-hour incentive?

Ops / Growth / Finance readout · 10-week randomised test (zone-day), window-months 16–18

The zone-hour incentive **works on marketplace liquidity** — but the flat per-delivery bonus **does not pay for itself**. Improve the design before rolling anything out.

Recommendation: do not roll out the flat bonus as tested. Restructure it to reward incremental supply, cap it to the supply-short and balanced zones, and re-run.

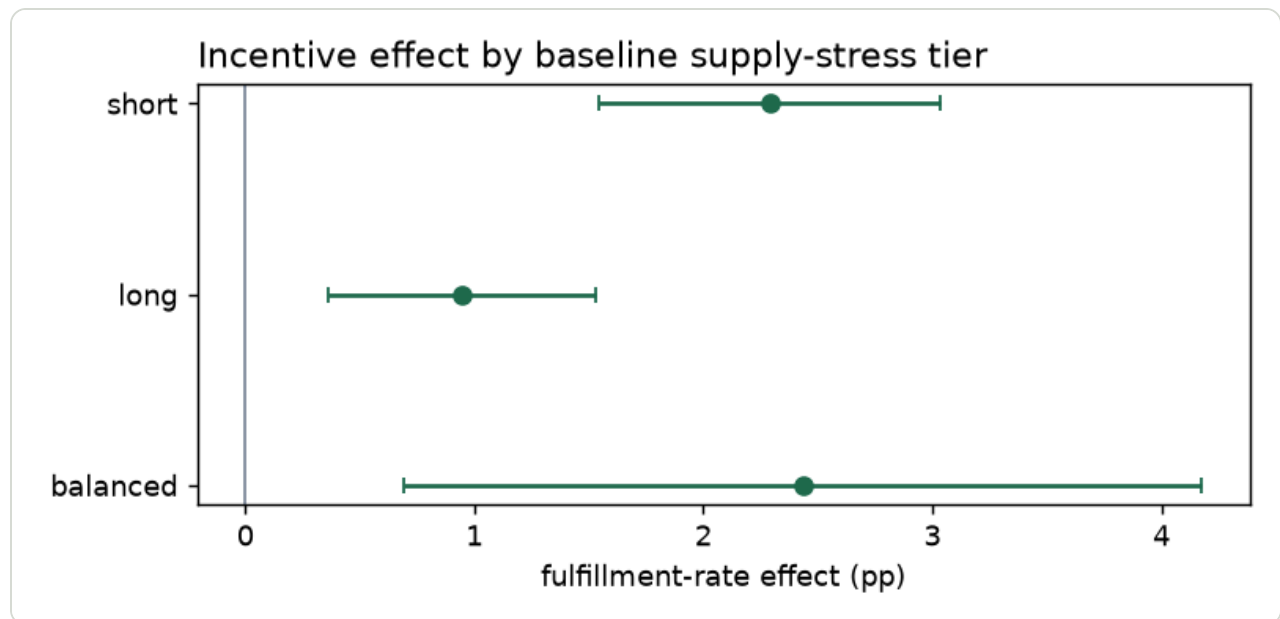
Did it work? Yes — on liquidity.

Metric (zone-day)	Control	Treatment	Effect	95% CI	Robust?
Fulfillment rate (primary)	95.7%	97.6%	+1.87 pp	[+1.27, +2.46]	cluster p 7e-10 · RI p 3e-4 · wild-boot agrees
ETA p90 (min) — co- primary	46.2	40.1	-6.06	[-8.56, -3.57]	all p < 0.002
No-courier cancel rate — co-primary	1.16%	0.80%	-0.36 pp	[-0.64, -0.08]	p ≈ 0.01–0.03
Courier earnings / active hour	R\$ 33.0	R\$ 41.2	+R\$ 8.12	[+5.6, +10.6]	p 1e-10

Where it works — supply-short and balanced zones, not the well-supplied ones

Baseline supply-stress tier	Zones	Fulfillment effect	95% CI
short	6	+2.29 pp	[+1.54, +3.04]
balanced	2	+2.43 pp	[+0.69, +4.18]
long (well-supplied)	4	+0.94 pp	[+0.36, +1.53]

Arm × tier interaction (cluster-robust Wald) $p = 0.004$; the short-tier and long-tier CIs do not overlap. In the long zones the bonus is **pure cost**.



Why not to ship it — the economics (guardrail G1)

Incentive spend (treated zone-days)	R\$ 35,820 on 7,446 delivered orders
Incremental delivered orders	≈ 127 — only 1.7% of the subsidised orders
Cost per incremental delivered order	≈ R\$ 281 (95% CI [R\$ 195, R\$ 431])
Contribution margin of a delivered order	≈ R\$ 10 → the bonus fails G1 by ~28×
Value destroyed per subsidised order	≈ R\$ 4.6

The bonus is a **flat payment on every delivery** in the treated zone-block, while ~98% of those orders would have been delivered anyway. The lever pulls real supply (+1.7 courier-hours per treated zone-day); its **pricing** is wrong.

Side effects — small

Neighbour-zone cannibalisation: control zones next to a treated zone fulfil ~ 0.26 pp lower ($p = 0.76$ — not significant); the per-treated-neighbour drag is -0.19 pp ($p = 0.42$). Implied $\sim 17\%$ of the local gain is supply pulled from neighbours $\rightarrow$ metro-level net $\approx 83\%$ of the summed local lift. Present, small, not statistically resolved at 12 zones. **Decay:** no detectable trend over the 10 weeks (the weekly series is too thin to confirm or rule out a launch bump).

Next steps. (1) Redesign the incentive to pay for *incrementality* — a threshold / surge-triggered payment that fires only when a zone is genuinely short, not a flat per-delivery bonus. (2) Scope any re-run to the supply-short + balanced zones only. (3) Keep the neighbour-zone spillover guardrail and a hard budget cap live. (4) Give a re-run more zone-days per week so the decay question can be answered.

Synthetic, seed-generated data (a portfolio project). Unit of randomisation = the zone-day (~ 840 , 420/arm, 12 zones); every headline number is computed on two tracks (pandas + DuckDB) with a parity assertion — `python data_analysis/parity/run_all.py` — and reported with cluster-robust, randomization-inference and wild-cluster-bootstrap inference. The true effect lives only in `etl/config.py::INCENTIVE_EFFECT` and was recovered from the data. Per-analysis detail: `data_analysis/findings/01 - 08`.